Why a Perfect AUC Is Insufficient: Bandwidth-Dependent Margin Loss in Ultra-Wideband Microwave Bone-Fracture Detection

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***Abstract*—Simulation studies of microwave fracture detection routinely report near-perfect classification. We show that such a figure can be uninformative, and that reporting it alone can support a materially wrong engineering decision. Using a full-wave finite-difference time-domain model of a layered forearm, we generated 192 ultra-wideband (3–11 GHz) acquisitions spanning eight simulated specimens, four soft-tissue overlay thicknesses and six fracture states, and scored each under three independent instrument-error realisations. Fractured and intact limbs separate by 40.8 dB with no overlap, giving an area under the receiver operating characteristic curve (AUC) of 1.0000. Because that statistic is saturated, we re-scored the identical 192 cells over restricted frequency windows, a controlled manipulation requiring no further electromagnetic simulation. Removing 3.00–4.35 GHz reduced the class separation from 40.8 dB to 17.9 dB, more than half, *while the AUC remained exactly 1.0000*. The metric registered none of the loss. We identify the mechanism as a loss of range resolution, 3.67 mm to 4.42 mm, which prevents migration from collapsing a layered-interface artefact and converts it into a specimen-dependent false alarm. We argue that any study reporting a saturated classification statistic should report the underlying margin in decibels alongside it, and we quantify what that practice would have changed here.**

***Index Terms*—microwave imaging, ultra-wideband, bone fracture, FDTD, detection margin, area under curve, model evaluation**

I. Introduction

Fracture diagnosis is dominated by projection radiography and computed tomography, both of which deliver ionising radiation and require fixed clinical infrastructure. This makes them poorly suited to repeated monitoring of healing, and unavailable at the point of injury. Microwave imaging is an attractive alternative because bone and the surrounding soft tissue differ substantially in permittivity, and because a fracture filled with blood presents a strong dielectric discontinuity against cortical bone [1].

The physical difficulty is that the water content creating the contrast also creates the loss. The interrogating signal must cross skin, fat and muscle twice, and attenuation rises steeply with frequency. The usable band is therefore a compromise between range resolution, which improves with bandwidth, and penetration, which degrades with it.

Simulation is the natural first instrument for exploring that compromise, and a large body of such work reports excellent discrimination between fractured and intact bone. This paper is concerned with how those results are *evaluated* rather than with whether they are achievable. Our starting observation is that a classifier which separates two classes completely produces an AUC of exactly 1.0000, and that this value is a *ceiling*. Once reached, the statistic cannot distinguish a system with an enormous margin from one that is close to failing. Any subsequent degradation, whether from reduced bandwidth, a harsher error model or a more difficult anatomy, is invisible until it becomes severe enough to cause an actual misclassification.

A. Evaluation practice, and where it fails

Classification metrics are the common currency of this literature: area under the curve, accuracy, and paired sensitivity and specificity. All three are bounded, and all three therefore saturate. Saturation is not a subtle property — a classifier that makes no errors attains the bound by definition — but its practical consequence is rarely exercised, because doing so requires deliberately degrading a working system and re-measuring.

The consequence is that above the saturation point these metrics report membership of a set, “makes no errors”, rather than a position within it. Two systems can both report an AUC of 1.0000 while one has an order of magnitude more margin than the other against the error sources its model omits. For a simulation study, where the error model is an assumption rather than a measurement, that distinction is the whole question: the margin is precisely the budget available to the terms the simulation does not contain.

This is familiar in another guise. Statistical practice has long distinguished significance from effect size, on the grounds that a p-value answers a different question from “how large is it”. The argument here is the same one applied to detection: a

TABLE I
SEPARATION OF FORWARD AND INVERSE MODELS

	Forward (data)	**Inverse (image)**
Physics	dispersive FDTD	homogeneous migration
Tissue	6 layers, Debye	one scalar ϵ_{avg}
Geometry	known kerf, cortex	not supplied
Grid	0.2 mm FDTD cells	independent grid
Error	injected	blind

classification statistic answers "are they separable", and once the answer is yes it cannot answer "by how much". What is reported alongside it should.

We make three contributions.

1) A full-wave study of ultra-wideband fracture detection comprising 192 acquisitions across a factorial design, analysed under a statistical plan fixed before the data were examined, including leave-one-specimen-out validation and a bootstrap that resamples specimens rather than measurements.
2) A controlled demonstration that a saturated AUC conceals the loss of more than half the detection margin. Because the manipulation is applied to the stored full-band response of the same solved cells, bandwidth is varied with everything else held exactly constant.
3) Identification of the mechanism, and a concrete reporting recommendation whose effect on our own engineering decisions we quantify.

II. METHOD

A. Tissue model

Permittivity and conductivity follow the Gabriel four-Cole–Cole parameterisation [1]. A time-domain solver cannot integrate a Cole–Cole response directly, because the fractional-power relaxation has no finite-order time-domain equivalent, so each tissue is refitted to a two-pole Debye model over 3–11 GHz. The worst deviation across all seven tissues is 0.23 %, and every relaxation time is checked to exceed the solver time step.

B. Phantom and experiment design

The phantom is a layered forearm: skin (2.0 mm), fat (3.0 mm), muscle (1.0 mm), near cortical wall (4.0 mm), marrow (10.0 mm) and far cortical wall (4.0 mm), with the antenna at a 12 mm standoff. A soft-tissue overlay of 0, 3, 6 or 10 mm spans anatomical variation. The fracture is a rectangular blood-filled kerf in the near cortex of width 0.2, 0.4, 1.0 or 2.0 mm, with an additional oblique state.

Eight specimens are drawn from a variation model perturbing cortical wall thickness and overlay, giving a factorial matrix of $8 \times 4 \times 6 = 192$ acquisitions. Without this between-specimen variation the leave-one-specimen-out analysis would be vacuous, because there would be nothing to leave out.

C. Forward model and the inverse crime

If the model generating the data and the model reconstructing the image share assumptions, reconstruction always succeeds and the result is circular. Table I lists the structural separations used to avoid this. The reconstructor receives a matrix of complex S_{11} values and nothing else: not the permittivity profile, the bone geometry, the fracture position or the noise realisation.

Data are generated with gprMax [2] on a 0.2 mm grid in two dimensions (TM_z). An independent analytic layered-medium model, fast enough for development, is used only to build and unit-test the processing chain; every result reported here comes from the full-wave solver.

D. Processing chain

Figure 1 shows acquisition through detection. The antenna steps along the bone axis through 61 positions at 1 mm spacing, transmitting and receiving at each, giving one reflection coefficient per position per frequency. Empty-scene subtraction removes the antenna response; a selective singular-value decomposition removes the layered clutter; Kirchhoff migration focuses the residual scattering back to its origin; and a scalar peak-over-noise metric scores the result in decibels.

One parameter required revision. A common recommendation is to discard the first two or three singular components. Measured on two independent targets, this destroys the fracture signature along with the clutter. We instead discard a component only if, after migration, its energy spans more than half the scan aperture, which is the signature of a layered interface rather than a localised scatterer.

E. Instrument error

Receiver noise, cable phase drift, stage jitter and standoff variation are injected *after* the forward solve. One electromagnetic solution therefore yields several independent "remounts", which is both cheaper and physically correct, since a remount perturbs the instrument rather than the anatomy. Three realisations per acquisition give 576 scores.

F. Statistical analysis plan

The plan was fixed before the data were examined. Two points require care. The detector has no fitted parameters, so the only trained quantity is the decision threshold; leave-one-specimen-out validation is therefore applied to the threshold rather than to the detector. The bootstrap resamples *specimens*, not measurements, because twenty-four acquisitions from one bone are not twenty-four independent observations and resampling them would give an interval several times too narrow. Permutation labels are shuffled within specimen for the same reason.

III. RESULTS

A. Detection performance

Table II gives the raw distributions. The classes are separated by **40.8 dB with no overlap**, a factor of 110 in amplitude. The AUC is 1.0000 at every overlay thickness

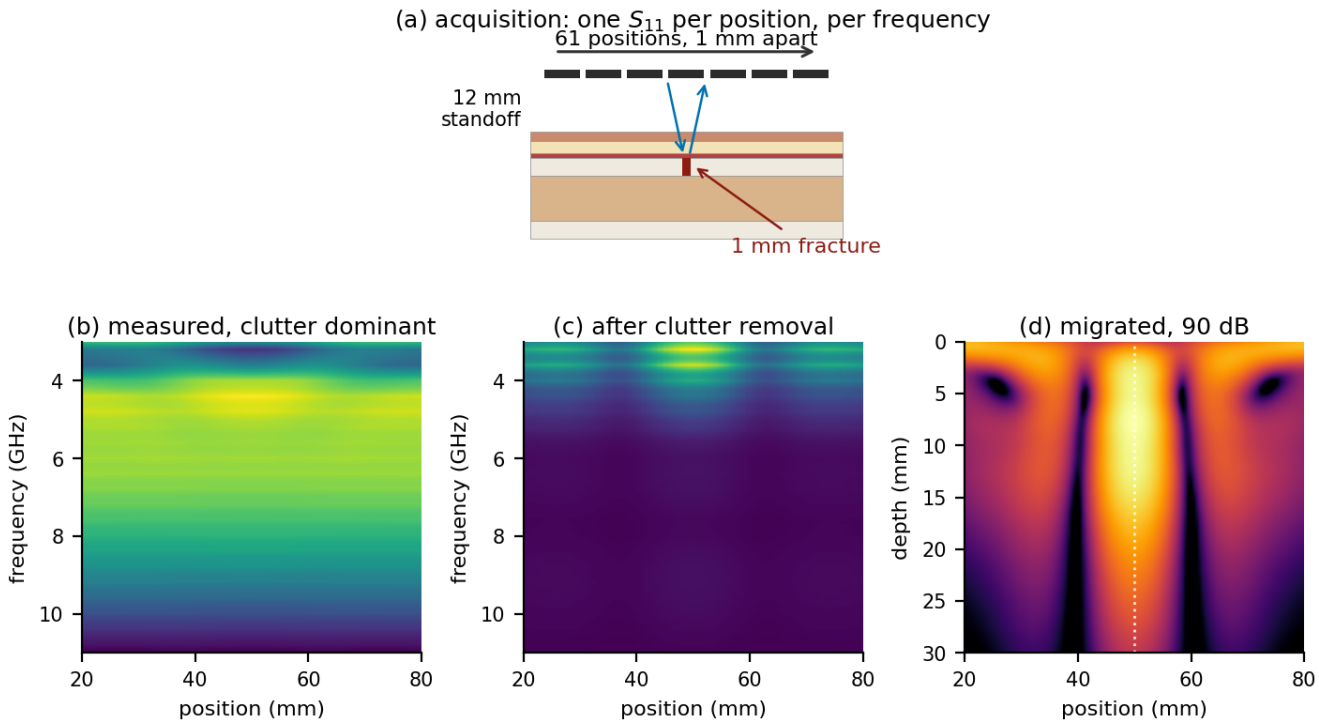


Fig. 1. Acquisition and processing. (a) The antenna steps along the bone axis at 12 mm standoff, transmitting and receiving at each of 61 positions. (b) The measured response after empty-scene subtraction, dominated by layered clutter; the fracture is present and invisible. (c) After selective singular-value removal, the local scattering remains. (d) Migration focuses it to the true position (dotted). Panels (b)–(d) are solver output for a 1 mm fracture under 6 mm of overlay.

TABLE II
DETECTOR SCORE DISTRIBUTIONS, 576 SCORES

Class	n	Min	Median	Max
Intact	96	8.8	19.3	**27.7**
Fractured	480	**68.5**	89.9	104.8

TABLE III
SEPARATION RESOLVED BY CONDITION

Condition	n	Weakest fract.	Separation
Soft-tissue overlay			
0 mm	144	88.9 dB	69.3 dB
3 mm	144	82.2 dB	61.3 dB
6 mm	144	81.0 dB	54.6 dB
10 mm	144	68.5 dB	**40.8 dB**
Fracture width			
0.2 mm	96	68.5 dB	**40.8 dB**
0.4 mm	96	72.0 dB	44.3 dB
1.0 mm	192	77.7 dB	50.0 dB
2.0 mm	96	82.1 dB	54.4 dB

and every fracture width, with a bootstrap 95 % interval over specimens of 1.000–1.000 and a permutation p of 0.001, the floor for 1000 shuffles. Leave-one-specimen-out validation of the threshold gives sensitivity 0.9979 (Wilson 0.988–1.000) and specificity 1.0000 (0.962–1.000). Repeatability across remounts is ICC(1,1) = 0.9979.

A perfect AUC ordinarily indicates leakage between training and test data rather than a genuine result, and was treated as such. Inspection of the distributions excludes that explanation: the gap is real, and its structure follows physical prediction. The weakest fractured cases are the narrowest kerf under the thickest overlay, and the strongest intact cases also occur at the thickest overlay. A leak would not order itself by physical difficulty.

B. Performance across the design space

Table III resolves the result by condition. Separation falls monotonically with overlay, from 69.3 dB with no overlay to 40.8 dB at 10 mm, and rises monotonically with fracture width, from 40.8 dB at 0.2 mm to 54.4 dB at 2.0 mm. Both orderings follow physical prediction: a thicker overlay attenuates the two-way path, and a narrower kerf presents a smaller dielectric discontinuity.

Two consequences follow. First, the headline 40.8 dB is the *worst case* over the entire design space, not a typical value; the median condition is considerably more comfortable. Second, the AUC is 1.0000 in every one of these eight strata, so the entire 28.5 dB spread visible in Table III is invisible to it. The oblique state, which presents an angled face and might be expected to scatter away from the antenna, gives a minimum of 78.2 dB, between the 1.0 and 2.0 mm transverse kerfs; obliquity does not penalise detection in this geometry.

TABLE IV
THE SAME 192 CELLS, RE-SCORED OVER RESTRICTED BANDS

Window	BW	Res.	AUC	Sep.
3.00–11.0 GHz	8.00 GHz	3.67 mm	1.0000	**40.8 dB**
4.35–11.0 GHz	6.65 GHz	4.42 mm	1.0000	**17.9 dB**
5.00–11.0 GHz	6.00 GHz	4.90 mm	1.0000	16.0 dB

TABLE V
FEED TAPER SWEEP, PORT FIXED AT 1.74 MM

Flare-end width	Worst	Covered	Re(Z_{in}) @ 4 GHz
1.74 mm (uniform)	−6.61 dB	84 %	24.4 Ω
2.75 mm	−8.50 dB	79 %	25.9 Ω
3.50 mm	−8.14 dB	94 %	29.1 Ω
4.00 mm	−9.80 dB	98.1 %	36.7 Ω
4.50 mm	**−9.98 dB**	**99.69 %**	**39.0 Ω**

C. Bandwidth-dependent margin loss

Every solved cell stores the full-band response, so the entire matrix can be re-scored over a restricted window with no further electromagnetic simulation. Bandwidth is thus varied with the geometry, the error model, the processing chain and the detector all held exactly constant.

Table IV and Fig. 2 give the result. Removing 3.00–4.35 GHz costs **22.9 dB, more than half the total separation**. The AUC is 1.0000 in all three windows and registers none of it.

D. Where the penalty falls

The aggregate figures of Table IV conceal a further structure. Fig. 3 resolves the loss by overlay thickness. It is −3.7 dB with no overlay, −10.1 dB at 3 mm, −25.4 dB at 6 mm and −15.5 dB at 10 mm.

The penalty is therefore *not* monotonic in difficulty, and is largest in the middle of the clinical envelope rather than at its hardest point. This matters methodologically: a study that characterised the cost of reduced bandwidth only at its worst-case operating point, which is the natural thing to do, would have recorded −15.5 dB and understated the true worst case by 10 dB.

E. Mechanism

Band limitation degrades both classes, and the intact class is the more damaging. The weakest fractured score falls from 68.5 to 55.7 dB, as expected from reduced penetration and resolution. The strongest *intact* score *rises* from 27.7 to 37.8 dB, which is a false alarm rather than a missed detection. It occurs on one specimen at 6 mm overlay against a group mean of 23.7 dB.

The mechanism follows from the resolution column of Table IV. Bandwidth buys range resolution. At full bandwidth, migration resolves that specimen's layered-interface reflection and collapses it; band-limited, the artefact smears into the detection window and registers as a target. Because the effect is specimen-dependent, the concern is variance rather than the mean shift: one specimen in eight spikes, and a larger cohort would be expected to contain worse outliers.

F. Consequence for the instrument

This measurement was not academic. An antipodal Vivaldi antenna [4] designed in openEMS [3] initially matched 4.35–12.00 GHz against a 3–11 GHz target, and the shortfall had been documented as an acceptable compromise on the strength of an unchanged AUC. Table IV shows that compromise cost 22.9 dB. Re-examining the input impedance showed the shortfall was never a low-frequency cutoff: the design was already matched at 2.225–2.725 GHz, reaching −33 dB, two gigahertz below its stated band edge. What lay between was a mismatch, a resistive plateau of 24–37 Ω across 3.25–4.40 GHz. A load of 24.4 Ω into a 50 Ω feed predicts −9.3 dB return loss and −9.14 dB was measured.

The remedy follows from the diagnosis. Widening the feed where it meets the flare, while holding the port at the 1.74 mm that presents 50 Ω to the connector, converts the feed into an impedance transformer. Table V shows the sweep: Re(Z_{in}) at 4 GHz rises monotonically with flare-end width, and in-band coverage follows. The first two entries read as a failure and the direction was initially judged wrong; the third reversed that judgement, which is recorded here because two points of a monotonic trend are not a trend.

The adopted 4.5 mm taper moved the match to 2.48–12.00 GHz, covering 99.69 % of the design band with a worst in-band value of −9.98 dB and a median input resistance of 45.6 Ω. Peak gain is 7.02, 11.87, 12.49, 12.06 and 9.33 dBi at 3, 5, 7, 9 and 11 GHz respectively; the 8–10 dBi target is met from approximately 4 GHz upward and is not met at 3 GHz, where the aperture is 0.6 wavelengths.

G. Exposure

The band straddles a boundary in current guidelines: specific absorption rate applies below 6 GHz and absorbed power density above it [5], [6]. Both were evaluated for a layered limb at 12 mm standoff, with tissue densities from [7]. At the specified 0 dBm transmit power and worst-case illumination, all limits pass with a minimum margin of 18×; the binding quantity is absorbed power density above 6 GHz.

IV. DISCUSSION

A saturated statistic is a lower bound, not a measurement. The defensible claim from Table II is that under the modelled error budget the classes separate by 40.8 dB with no overlap, not that detection is perfect. The binding assumption is the error model rather than the physics: a physical bench adds cable flex, connector repeatability, operator variation and genuine inter-specimen heterogeneity, none of which are simulated. The 40.8 dB margin is the budget those unmodelled terms must consume before the conclusion changes, and that framing is only available because the margin was reported.

The practical consequence is concrete. Judged by AUC alone, the initial antenna was adequate: the statistic did not move. Judged by margin, it cost more than half the available

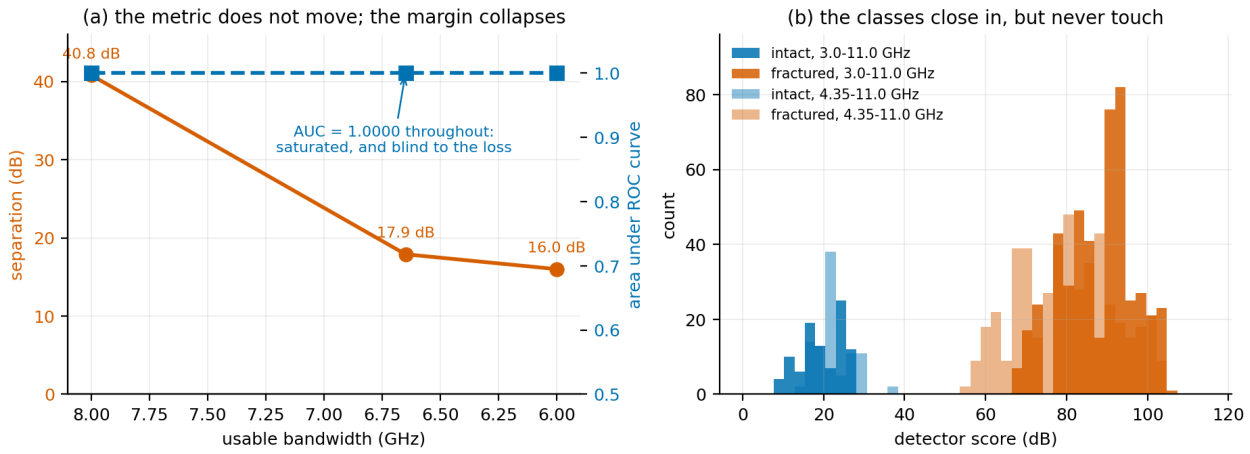


Fig. 2. (a) As usable bandwidth falls, the class separation collapses from 40.8 dB to 16.0 dB while the AUC remains pinned at 1.0000. (b) The underlying distributions converge from both sides but do not overlap, which is precisely why the classification statistic cannot see the change.

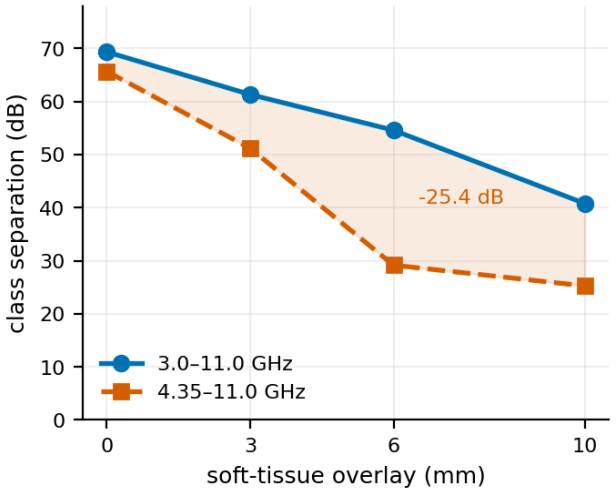


Fig. 3. Separation against soft-tissue overlay at full and restricted bandwidth. The penalty is not uniform: it is largest at 6 mm, where the band-limited false alarm appears, rather than at 10 mm where the signal is weakest. Measuring the penalty only at the hardest operating point would understate it by 10 dB.

budget, which is what motivated the redesign reported above. The same reasoning applies to any design decision evaluated against a saturated metric, including grid refinement, averaging time and dynamic range.

We therefore suggest that studies reporting near-perfect classification also report the following, none of which requires an additional experiment.

1) *The separation, in the detector's own units*, together with the extreme values of each class. Table II is four numbers and carries more information about this system than the AUC does.
2) *The separation resolved by condition*, not only in aggregate. Table III shows a 28.5 dB spread across strata that all report an identical AUC, and Fig. 3 shows that a degradation can fall hardest in the middle of the operating envelope rather than at its edge.
3) *One deliberate degradation*, re-measured. Restricting the band cost nothing here because the stored responses were full-band, and it converted an assumption about the instrument into a number.

The third is the one we would most encourage. A simulation study can almost always degrade itself cheaply — bandwidth, dynamic range, averaging, grid refinement — and the resulting slope is far more informative than any single operating point. Reporting only the operating point is what allows a saturated metric to pass unnoticed.

A. Reproducibility

Every figure and every numerical value in this paper is regenerated by a script from stored solver output; none is transcribed by hand. The analysis code, the deck generators, the statistical plan and the figure sources are public, and the statistical plan was committed before the matrix was complete.

V. Limitations

1) The injected error model is the binding assumption. A physical bench adds terms this work does not model.
2) Every fracture is at a known position; the study measures separability, not localisation.
3) Propagation is two-dimensional TM_z. A three-dimensional cylindrical model has been implemented and verified against the solver but the comparison is not yet run, so no claim is made about the effect of curvature or out-of-plane scattering.
4) Eight simulated specimens are not eight cadaveric bones.
5) The imaging model excites with an elementary dipole, not the antenna of Section III-D; antenna and imaging results are independent.
6) Antenna gain at 3 GHz is unmet, and was shown not to be reachable by resizing within the specified board.
7) Exposure is assessed by calculation, which is not a substitute for measurement to IEC 62209.

No hardware was built and no human or animal subject was exposed at any point in this work.

VI. Future Work

Four steps follow directly, in the order we judge most informative.

Measure the limit rather than bound it. Every result here is a lower bound: the classes never overlap, so the experiment reports that the separation exceeds what it can resolve. Pushing until detection fails would convert the bound into a measurement. The cheapest lever is a harsher error model, the next a thicker overlay, and the most expensive a kerf finer than the 0.2 mm the present grid can represent.

Randomise the fracture position. Every fracture here is at a known location, so the study measures separability and not localisation. Drawing the position uniformly along the scan and scoring localisation error would test the migration step rather than assuming it.

Complete the three-dimensional comparison. A cylindrical limb model with the cortex offset toward the skin, so that the near wall sits at the same 6 mm depth as the layered phantom, has been implemented and verified against the solver; the 192-cell equivalent has not been run. Three effects act in opposition — a curved surface spreads the reflected wavefront, energy leaves the plane that a two-dimensional model traps, but a transverse crack through a curved wall presents a longer scattering edge illuminated over a range of incidence angles — so the sign of the net effect is a measurement rather than an argument.

Validate against measurement. The error model is the binding assumption throughout, and only a physical bench can calibrate it. A machined phantom with a known kerf, measured on a vector network analyser, would convert the dominant assumption of this study into data.

VII. Conclusion

Across 192 full-wave acquisitions from eight simulated specimens, fractured and intact limbs separate by 40.8 dB with no overlap, giving an AUC of 1.0000 with leave-one-specimen-out sensitivity 0.998 and specificity 1.000. Restricting the usable band removed more than half of that separation while the AUC remained exactly 1.0000, and the mechanism is a loss of range resolution that converts a resolvable layered-interface artefact into a specimen-dependent false alarm. A perfect classification score is therefore a ceiling rather than a result, and should be accompanied by the margin it conceals.

All code, data-generating scripts and figures are available at https://github.com/amalnathkp13-boop/bone-fracture-rf-sensing.

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